

MA 301
Test 2
Fall - 2013

Problem	points	out of
1		10
2		10
3		14
4		5
5		5
Score		44

Name: _____

HR: _____

Signature: _____

Problem 1: Suppose $\mathbf{F} = [5z + 2xz, 1, x^2 + y]$ and S the disc: $x^2 + z^2 \leq 4$ in the plane $y = 1$ with orientation pointing away from the origin. Let C be a curve which is the boundary of S .

(a) Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using Stokes' Theorem

(b) Compute $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ directly

Problem 2: Let $\mathbf{F} = y \mathbf{i} + (2 - 3yz) \mathbf{j} + (x^2 + y^2) \mathbf{k}$. [Set-up](#) (but DO NOT compute) the integrals you would need to compute the following: (Limits of integration must be included)

(a) $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is a curve which bounds the quarter cylinder $x^2 + z^2 = 16$ with $x \geq 0, z \geq 0$, & $0 \leq y \leq 2$ and has counter-clockwise orientation.

(b) $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ where S is the surface of the solid bounded between $z = -x^2 - y^2 + 1$ and $z = 0$ with outward pointing normal.

Problem 3: Let $z_1 = 1 - i$, $z_2 = 2i$. Compute the following and leave answer in standard form.

(a) $z_1 - z_2$

(b) $z_1 z_2$

(c) $|z_2|$

(d) $\frac{z_2}{z_1}$

Solve the following complex equations and leave your answer(s) in Euler's Form: $re^{\theta i}$. No Decimals.

(e) $x^{1/5} + z_2 = 0$.

(f) Solve for x : $\frac{x^3}{z_2} - z_1 = 0$. (explicitly write out multiple values)

Problem 4: Let S be a surface given by $z = x^3y$ where $0 \leq x \leq 5$ and $-1 \leq y \leq 6$. **Set-up** (DO NOT Compute) the integral you would use to compute the surface integral $\iint_S (1 + 9xz)^{3/2} dS$

Problem 5: Let $\mathbf{F} = [x, ye^z, e^z]$ and S be the surface (with outward orientation) of the box $|x| \leq 1$, $|y| \leq 2$, $0 \leq z \leq 1$.

Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dA$